HOSPITAL READMISSION RATE ANALYSIS

Linking CMS hospital performance with county-level socioeconomic context

ANALYTICAL FOCUS	METHOD	DATA
Excess readmission ratios	**Linear mixed-effects model**	**CMS HRRP + HRSA AHRF**

Prepared by Khanh (Ada) Nguyen
Applied Statistics & Data Science | R | Data Integration | Healthcare Analytics

Executive summary

This portfolio study evaluates variation in CMS excess readmission ratios (ERR) across six Hospital Readmissions Reduction Program measures and examines whether county socioeconomic conditions remain associated with performance after accounting for repeated observations within hospitals.

7,944	2,444	1,197	55%
Hospital-condition observations	Hospitals	Counties	ERR at or above 1

Key adjusted findings

Adjusted association	ERR change per 1 SD	95% CI	Interpretation
Poverty	+0.0082	+0.0049 to +0.0116	Positive, statistically supported association
Unemployment	+0.0069	+0.0045 to +0.0094	Positive, statistically supported association
County population (log)	+0.0060	+0.0034 to +0.0086	Positive, statistically supported association

Takeaway Readmission performance is associated with local socioeconomic context, while substantial hospital-level variation remains after adjustment.

Data engineering and quality control

Three public sources were integrated: CMS HRRP condition-level measures, the CMS hospital directory, and HRSA Area Health Resources Files. County FIPS was assigned from hospital state and county name, with a conservative unique-ZIP fallback.

CMS HRRP		Eligibility filters		Hospital + county joins		Final model set
18,510 raw rows	→	**8,121 valid rows**	→	**8,121 unique keys**	→	**7,944 rows**

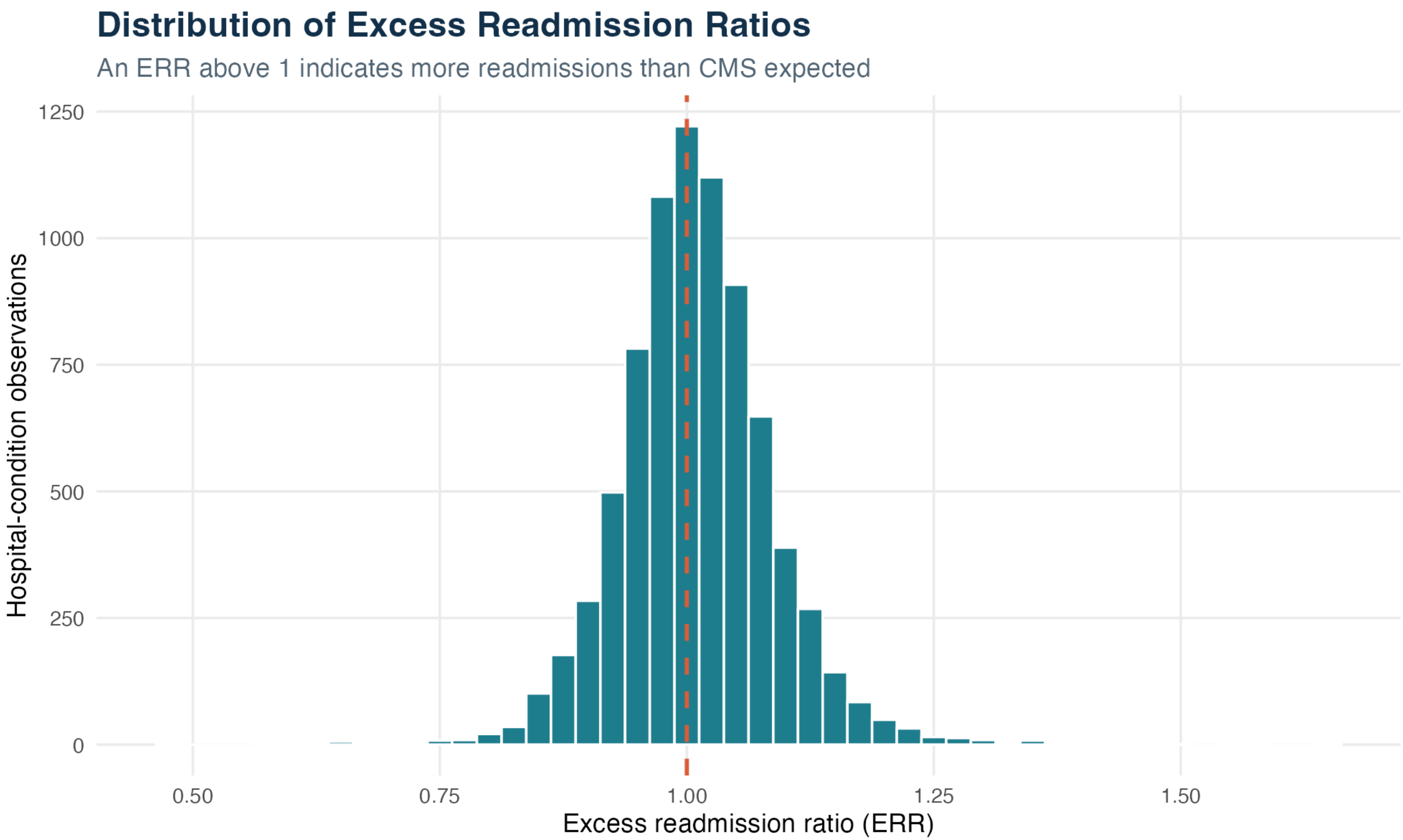
Critical correction

The original final dataset contained **2,694 duplicated hospital–measure keys**. The cause was a many-to-many ZIP-to-county join: one ZIP can cross county boundaries. The corrected pipeline uses county name plus state as the primary key and accepts ZIP only when it maps to a single county.

Validation checks

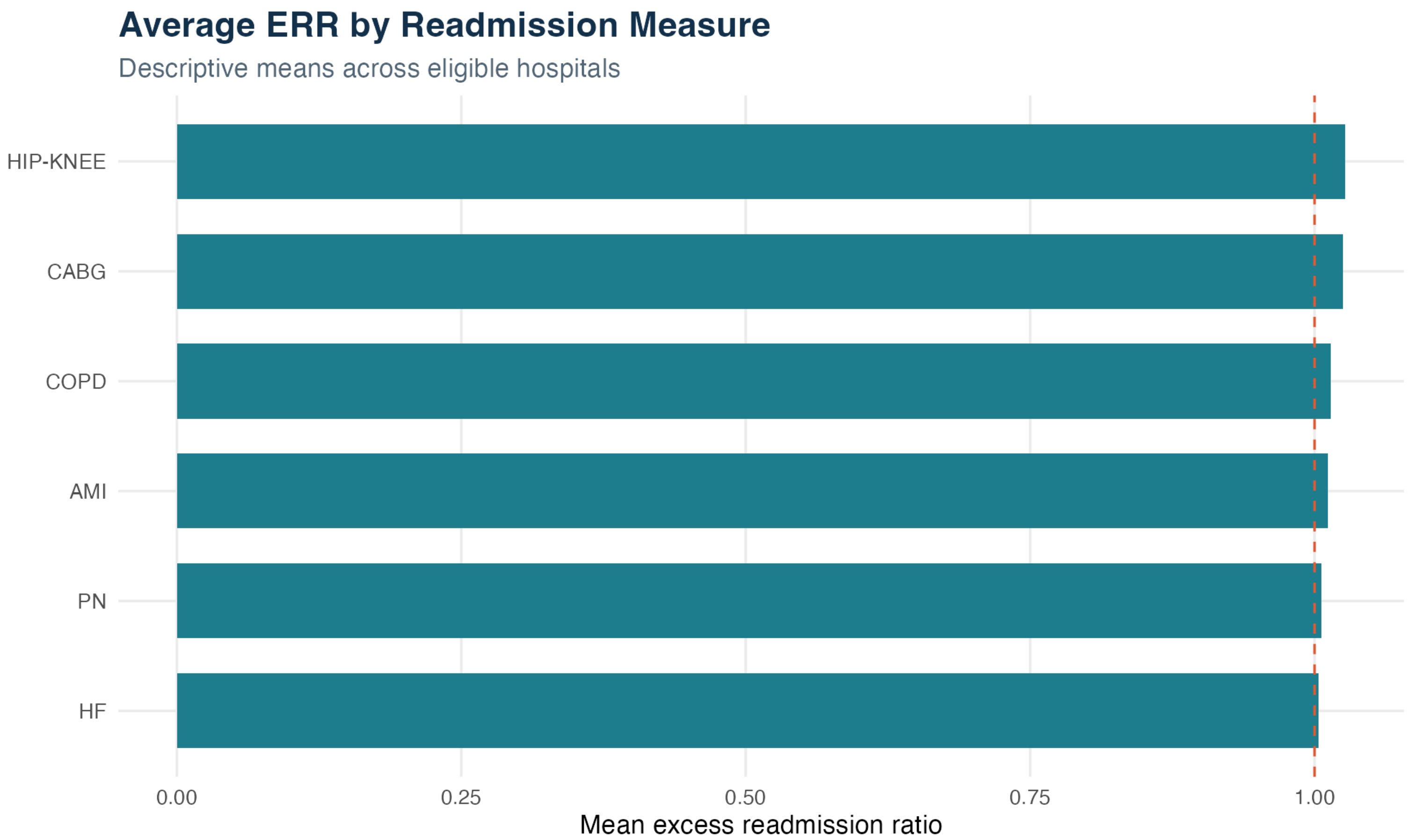
Check	Result
Hospital–measure uniqueness	0 duplicate keys after corrected joins
Row preservation	8,121 valid source rows → 8,121 joined rows
County FIPS coverage	8,035 of 8,121 rows (98.9%)
Complete socioeconomic records	7,946 rows (97.8%)
Final mixed-model analysis set	7,944 rows after excluding an unstable 1-hospital ownership group
Identifier handling	CMS facility ID retained as a six-character string

Outcome distribution



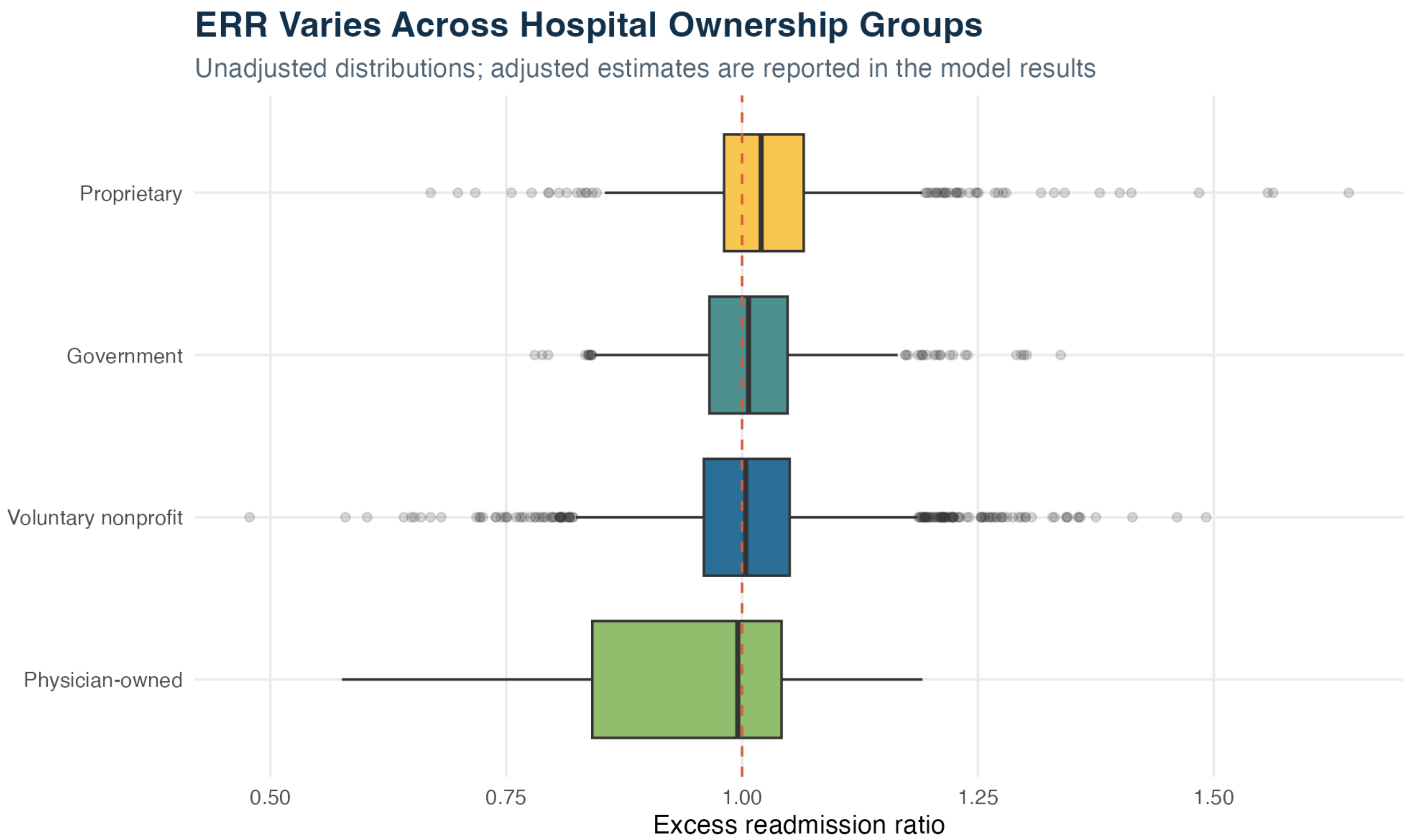
Interpretation	ERR centers close to 1, but the distribution includes meaningful tails. Descriptive comparisons should therefore be paired with adjusted modeling.

Variation by condition



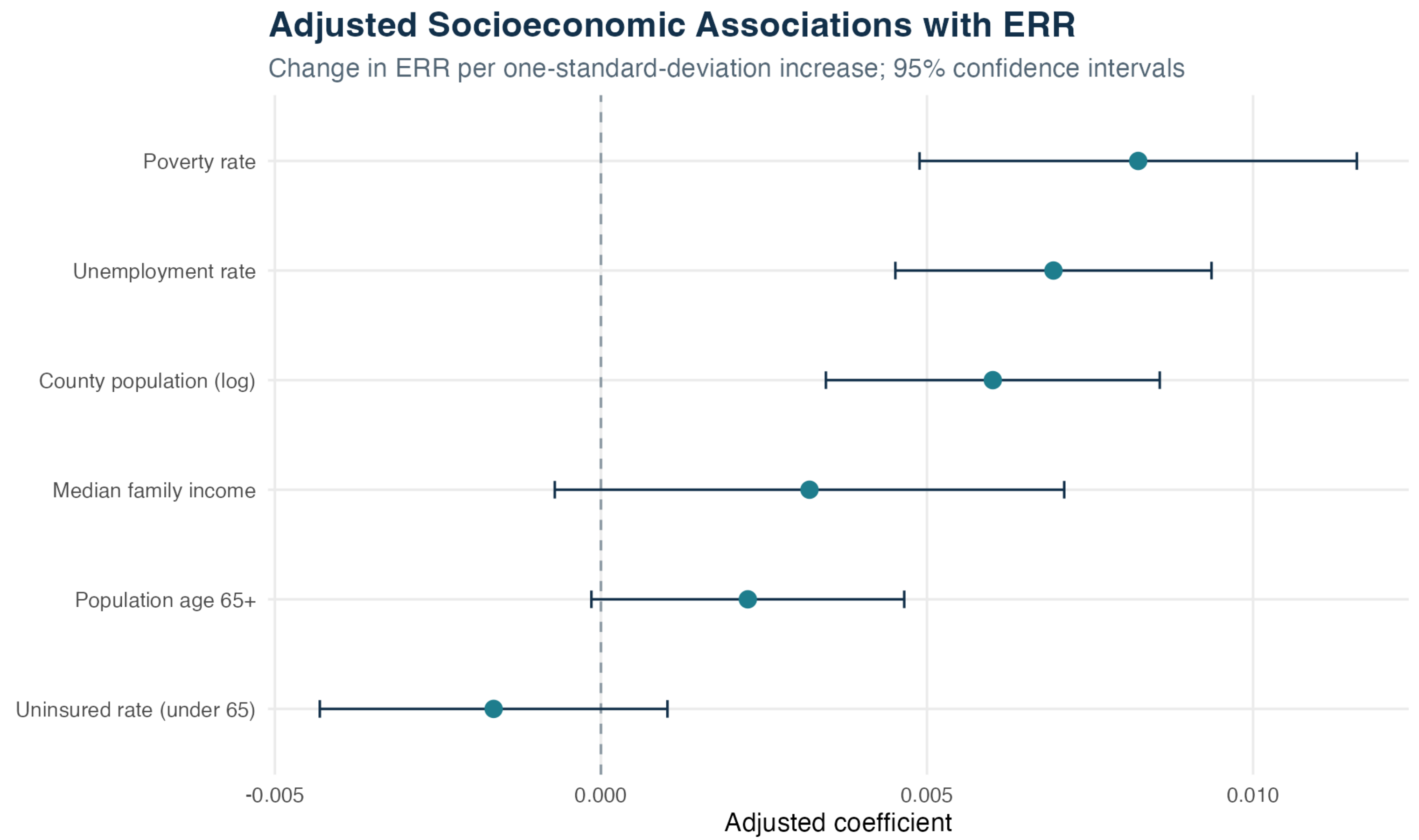
Interpretation	Average ERR differs across the six HRRP measures. Condition is included as a fixed effect so socioeconomic comparisons are not driven by the mix of measures.

Hospital ownership



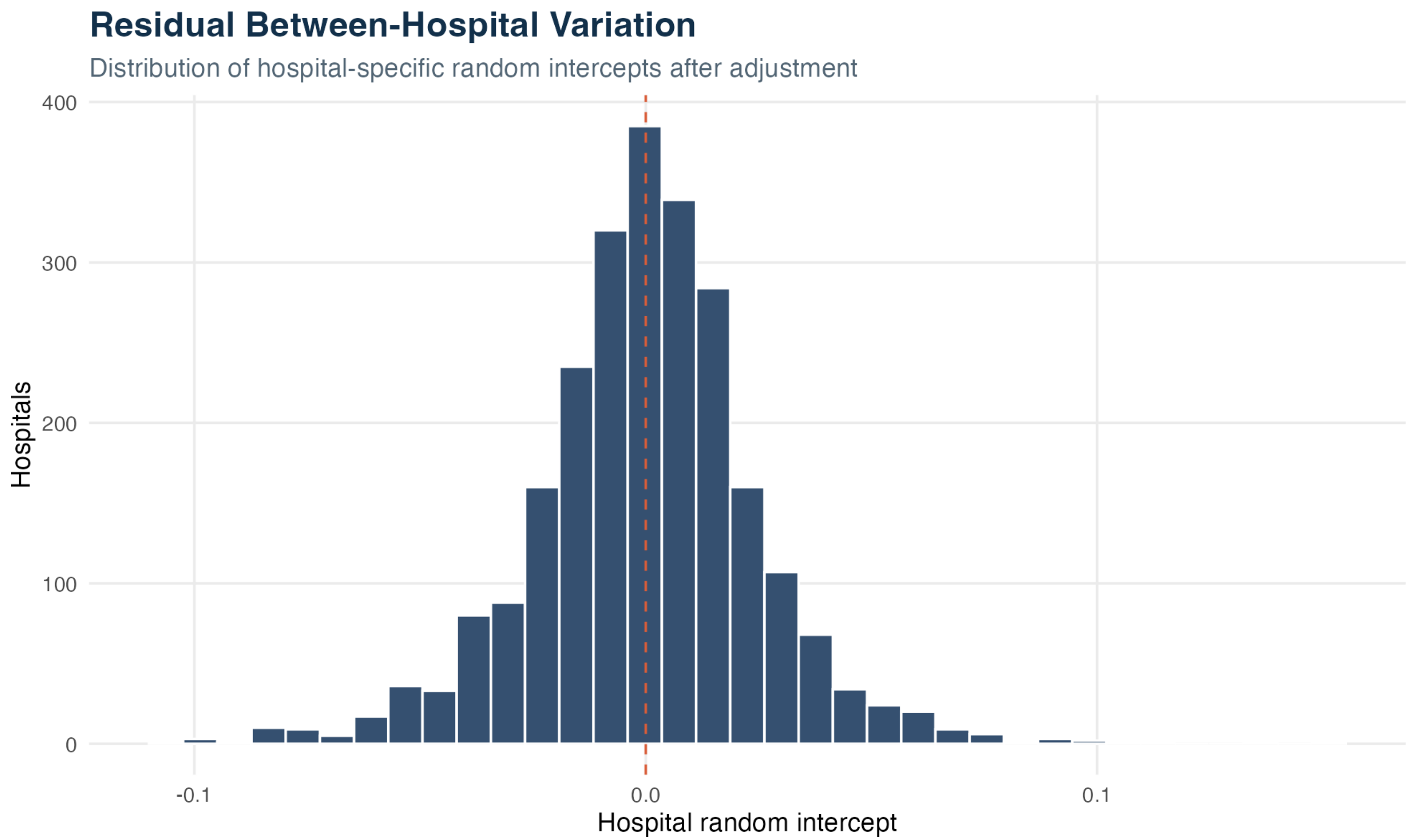
Interpretation	Unadjusted distributions overlap. In the adjusted model, physician-owned hospitals had lower ERR than voluntary nonprofit hospitals; interpretation is cautious because this category included only 26 hospitals in the complete model set.

Adjusted socioeconomic associations



Interpretation	Poverty, unemployment, and county population showed positive adjusted associations. Income, age 65+, and uninsured rate did not have confidence intervals fully separated from zero.

Between-hospital heterogeneity



Interpretation	The estimated intraclass correlation was 23.0%, indicating that repeated condition-level ERR observations from the same hospital share meaningful residual similarity.

Methods, limitations, and Tableau blueprint

Model specification

A linear mixed-effects model estimated ERR as a function of readmission measure, grouped ownership, emergency-service availability, and six standardized county covariates. A hospital random intercept accounted for repeated measures within provider.

Model improvement

Adding hospital characteristics improved fit over a condition-only model (likelihood-ratio $p<0.0001$). Adding socioeconomic covariates provided further improvement ($p<0.0001$).

Limitations

This is an observational, ecological analysis. County characteristics do not represent individual patients, associations are not causal, data years differ across sources, and some hospitals lacked complete county covariates. ERR is already risk-adjusted by CMS, but residual confounding remains.

Recommended Tableau views

1. KPI strip: hospitals, observations, counties, percent ERR ≥1
2. Condition comparison: mean ERR and observed rate
3. Ownership comparison with minimum-count disclosure
4. County socioeconomic scatterplots with filters
5. U.S. map using county FIPS
6. Tooltip definitions for ERR and every SES measure

Suggested dashboard question

Where do hospital readmission outcomes remain elevated, and how do hospital structure and county socioeconomic context relate to that variation?

Data package

Use **hospital_readmission_tableau_ready.csv** for flexible exploration or **tableau_model_complete.csv** when every view must use the same complete-case population as the statistical model.

Sources

Centers for Medicare & Medicaid Services, Hospital Readmissions Reduction Program: https://data.cms.gov/provider-data/dataset/9n3s-kdb3
Health Resources & Services Administration, Area Health Resources Files: https://data.hrsa.gov/data/download?data=AHRF

Analysis reflects the supplied CMS performance period 2020-07-01 through 2023-06-30 and supplied AHRF variables from 2021–2023.